

Wonder and Grow Healthy : User-Centered Design for Selective Eating Children with Autism Spectrum Disorder to Support Healthy Eating Habits

Parvaneh Parvin*
Human Nutrition and Health
Wageningen University and Research
Wageningen, Netherlands
parvaneh.parvin@wur.nl

Sanne Boesveldt
Human Nutrition and Health
Wageningen University and Research
Wageningen, Netherlands
sanne.boesveldt@wur.nl

Emilia Barakova
Industrial Design
Eindhoven University of Technology
Eindhoven, Netherlands
e.i.barakova@tue.nl

Abstract

Selective eating in children with autism spectrum disorder poses unique challenges due to sensory sensitivities, often leading to nutritional deficiencies and stress for families. Existing interventions are effective but resource-intensive and inaccessible due to clinics' long waiting periods. This study uses participatory design principles, engaging parents in the co-creation of a sensory game-based platform. Through focus group and thematic analysis, key features were identified, including tools for tracking sensory behaviors, progress milestones, and personalized expert guidance. For children, parents prioritized gradual, structured sensory exposure, motivational rewards, and hybrid game-based environments blending digital and physical interactions. Open discussions highlighted the potential of a user-centered digital platform to complement therapies aligning with children's evolving needs while supporting parents in managing mealtime challenges. The study culminates in a design brief proposing a home-based solution tailored to families' needs. Future work involves refining, prototyping, and usability testing to ensure its practicality and effectiveness in real-world settings.

CCS Concepts

• **Human-centered computing** → *User centered design*; **Participatory design**.

Keywords

Autism Spectrum Disorder; Picky Eating; Sensory Based Intervention; Game-based Platform; Parental Support, Focus Group

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*Also with Eindhoven University of Technology.

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1 Introduction

Selective eating, defined as the reluctance to consume a wide variety of foods based on sensory traits such as texture and color, is a common challenge among children with Autism Spectrum Disorder (ASD) [1, 5]. This behavior often leads to Avoidant/Restrictive Food Intake Disorder (ARFID) [17, 22], resulting in nutritional deficiencies that increase the risk of immediate and long-term health problems, including obesity, anorexia, and malnutrition [4]. Children with ASD are up to five times more likely to experience feeding difficulties than their typically developing peers [23], driven by heightened sensory sensitivities, developmental stages, and psychological factors [7, 10]. These sensitivities often trigger adverse reactions to specific food textures, flavors, and aromas, further limiting dietary diversity and nutritional adequacy. While interventions like Applied Behavior Analysis (ABA) [3] and Occupational Therapy (OT) [2] provide relief, they are resource-intensive and time-consuming. Parental involvement is key in approaches like ABA, where parents gradually introduce new foods through food chaining [11]. While this approach helps ease anxiety around new foods, it demands consistent parental engagement, placing considerable time and effort on families. However, professional support from clinics is inaccessible to many families due to long waiting times and high costs. In the Netherlands, clinic waiting periods range from 8 to 12 months, leaving parents to manage their children's eating challenges largely on their own. Approximately 25-40% of all young children in the Netherlands experience eating problems, and eventually, 3-5% are diagnosed with ARFID [24]. There is a significantly large group of children who require such therapies. Therefore, the waiting times build up and may vary from a few months to a year. These delays in accessing treatment force parents to bridge the gap independently, often without adequate support, leading to emotional and logistical strain. Thus, there is a pressing need for an accessible, adaptable, and parent-friendly solution that addresses sensory and behavioral challenges without overwhelming families. This study adopts participatory design principles to co-create a digital tool tailored to the dietary and sensory needs of children with ASD while supporting their parents. Through focus group discussions, it explores parents' perspectives on their children's selective eating behaviors, evaluates existing interventions, and collaboratively defines design requirements for a game-based tool for children and a companion app for parents. Since children with ASD often struggle to articulate their experiences, parents play a key role in providing insights into their challenges and preferences. The findings contribute to Human Computer Interaction (HCI) research by offering empirical insights

into professional and parental strategies for managing selective eating in children with ASD and by informing the design of digital interventions that address both children's sensory and dietary needs while supporting their parents. The paper is structured as follows: it begins by demonstrating how participatory design can be leveraged to develop dual-user digital interventions that support both children with ASD and their caregivers. It provides empirical insights into effective and ineffective parental strategies for managing selective eating, offering guidance for interaction design in sensory-based interventions. Additionally, it bridges research on gamification and sensory-friendly interaction design to inform more adaptive and behaviorally-informed assistive technologies.

2 Literature Review

Research indicates that children with ASD often experience heightened sensitivity to sensory characteristics of food, such as texture, taste, and color, which contributes to selective eating patterns and ARFID [1, 6, 19]. Addressing selective eating in children with ASD [4] and associated feeding challenges such as ARFID requires a multidisciplinary approach that considers sensory, behavioral, and contextual factors. Traditional interventions such as ABA [3] and OT [2] are widely used to address these challenges. ABA incorporates structured techniques such as positive reinforcement, graduated prompting, and food chaining to reduce food-related anxieties and expand dietary diversity [11]. Similarly, OT focuses on sensory integration, helping children process sensory inputs and manage aversions to textures and tastes [20]. Similarly, methods such as Sequential Oral Sensory (SOS) and sensory diets [21] offer structured exposure to new foods, promoting sensory tolerance and familiarity. These strategies highlight the importance of tailoring interventions to individual sensory profiles for greater effectiveness. Although these approaches are effective, they are often time-consuming and resource-heavy and require significant parental involvement, placing substantial demands on families. Other interventions, including Cognitive Behavioral Therapy (CBT) [25] and pharmacological treatments [16], address social and emotional factors tied to selective eating. CBT can reframe food-related fears and improve social comfort during mealtimes. However, it may have limited effectiveness in addressing core sensory challenges [25]. Pharmacological treatments, target behavioral symptoms in ASD, but lack evidence for improving food selectivity [16]. Similarly, while nutritionists play a crucial role in designing balanced meal plans, they typically do not address the underlying behavioral and sensory factors contributing to selective eating.

The limitations of traditional approaches, including high resource demands and limited adaptability, have led to the exploration of technology-driven interventions to manage selective eating in ASD. Digital platforms using gamification offer structured and interactive environments tailored to individual sensory profiles [14]. These platforms allow children to keep track of their food choices while giving parents a way to monitor their child's progress, fostering independence in children, and providing real-time feedback to parents. [9, 12]. Game-based tools such as the Food Dudes program [13] while not specific to ASD, have successfully increased fruit and vegetable consumption through reinforcement and peer modeling strategies. Its success highlights the broader applicability

of gamified approaches to promoting healthy eating habits. LEGO therapy is another gamified approach that uses collaborative play, positive reinforcement, and task division to encourage engagement and independence [15], although developed primarily for social skill training. However, while these interventions successfully leverage gamification and reinforcement mechanisms, they often lack adaptation for ASD-specific sensory needs and do not fully engage parents as co-creators in the design process. Existing tools primarily focus on tracking and behavioral reinforcement but may not sufficiently address the nuanced sensory challenges that drive food refusal in children with ASD. This study extends prior research by integrating sensory-based gradual exposure, hybrid digital-physical interactions, and adaptive engagement strategies informed by parental experiences, ensuring a more tailored and sustainable intervention for both children and their caregivers.

3 Methods

The current study employs participatory design principles to ensure the intervention is user-centered and contextually relevant. This approach involves working closely with parents of children with ASD who experience selective eating behaviors to co-create practical solutions. An interactive focus group was selected as the primary method for qualitative data collection to gather information about parents' experiences, challenges, and preferences. Two focus group sessions were conducted online to ensure that the group size in each session remained small enough to encourage in-depth discussion and meaningful interaction among participants. Both sessions followed an identical process: 1) pre-session questionnaire: Before the session, participants completed a "Demographic and Use of Technology" questionnaire (See Appendix A.1). This provided background information on their use of technology, comfort levels with digital tools, and their children's eating behaviors; 2) focus group sessions: both sessions followed the same format, including structured discussions, open feedback rounds, and dot-voting activities (see Appendix B); and 3) post-session questionnaire: after the focus group, participants completed an "Expectation Scale for a Sensory Game-Based Tool and Companion App" questionnaire (See Appendix A.2). This included Likert-scale questions and open-ended questions.

3.1 Participants

The focus group participants included parents (all women) of children with ASD who also experience selective eating. Participants were recruited through SeysCentra, a center of treatment and expertise specializing in severe eating problems, including ARFID, for children and adolescents, as well as an ARFID Facebook page. All participants provided their informed consent before the study began. Table 1 summarizes the demographic characteristics of parents and their children along with the use of technology.

3.2 Focus Group Procedure

The two focus groups (N = 6) were conducted online via Microsoft Teams each lasting 120 minutes. To ensure participant engagement and thorough exploration of the topics, the session was divided into three structured rounds (Structured discussions; Open feedback; and Dot-voting activities;) with breaks. The researchers took on

Table 1: Demographic Characteristics of Participants and Their Children. In the Dutch education system, *MBO* (Middelbaar Beroepsonderwijs) refers to secondary vocational education, while *HBO* (Hoger Beroepsonderwijs) is a higher professional education comparable to a Bachelor's degree and *WO* (Wetenschappelijk Onderwijs) refers to university-level academic education.

Characteristic	Details (Number or Mean $\pm$ SD)
Parents' Age	Mean $\pm$ SD: 36.7 $\pm$ 7.5 years
Child's Age	Mean $\pm$ SD: 7.2 $\pm$ 1.9 years
Mother's Education	High school or below: 0, MBO: 3, HBO/WO: 1, Master or higher: 2
Work Status	Full-time: 0, Part-time: 4, Unemployed: 1, Other: 1
Marital Status	Married: 2, Single: 1, Separated: 2, Other: 1
Child's Eating Behavior	Very picky: 6
Uses Digital Tools	Yes: 4, No: 2
Comfort with Technology	Very comfortable: 1, Comfortable: 2, Neutral: 2, None: 1

different roles: one guided the discussions, another monitored the time, and a third recorded key concepts on a shared Miro board for visual support. In the first round, the discussion explored parents' challenges and experiences related to their children's selective eating behaviors, focusing on specific mealtime scenarios and the impact of sensory characteristics such as textures, smells, and tastes. The second round gathered feedback on existing interventions and strategies, with participants evaluating their effectiveness and limitations. The third round involved a dot voting activity to prioritize potential features of the proposed solutions. Parents assigned colored dots to predefined options for a parent app and a child's game, identifying the features they found most important. The round concluded with an open discussion, in which participants reflected on the ideas generated in this round. This method not only provided quantitative data on feature preferences but also encouraged participants to offer qualitative insights by explaining their choices.

3.3 Data Analysis

The interactive focus group sessions were audio-recorded and transcribed to capture the rich qualitative data shared by parents. As the focus group was conducted in Dutch, the audio recordings were first translated into English using "Otter.ai" software for further analysis. The qualitative data analysis followed a thematic approach. This involved a detailed review of the full transcripts, identifying and labeling key quotes related to the research questions as codes. These codes represented core ideas and recurring patterns in the parents' experiences, challenges, and preferences. These codes were then organized into sub-themes and overarching themes using an affinity diagram, which grouped similar concepts into logical clusters. This systematic process provided a structured framework for understanding the qualitative data and ensuring the results were grounded in the participants' own words. To further enhance the analysis and ensure the information was actionable for the design process, additional designer-specific methodologies were used. For example, data from Round 1, which focused on parents'

main challenges with picky eating behaviors, were synthesized into empathy maps for both the child and the parent. These maps captured emotional and cognitive aspects that might not be immediately evident in the raw data, offering a nuanced understanding of the participants' perspectives. Round 2, which centered on existing interventions and strategies, provided findings on effective and non-effective strategies. In Round 3, parents provided their input on the specific features and priorities for the proposed digital solution. Using dot-voting- "a technique that allows an agile team to quickly select or prioritize items with input from all team members." [8], participants ranked predefined features for both the parent app and the child's game. This exercise generated both quantitative data on feature prioritization and qualitative insights from the participants' justifications for their choices. The results of this round were analyzed to understand how parents valued different aspects of the proposed solution and how these preferences aligned with their lived experiences. To analyze the second questionnaire, Likert-scale questions were grouped into themes based on their content and intent of questions. Questions under each theme were aggregated, and their mean and standard deviation (SD) were calculated to provide a quantitative summary.

4 Results

The findings of the focus groups are structured into three themes: challenges and experiences, effective and ineffective strategies, and priorities for new solutions. Differences between the two groups are highlighted where relevant.

4.1 Challenges and Experiences

Participants from focus groups reported their experiences and significant challenges in managing selective eating behaviors. Three recurring themes emerged as follows.

- **Parental Control:** Parents described the burden of managing control of their child's eating habits while respecting their boundaries. Constant planning and stress often strained family dynamics, creating a tense mealtime atmosphere. *"You're basically planning all day"* – P2.
- **Food Refusal and Sensory Sensitivities:** Children exhibited strong preferences for "safe" foods, often rejecting new foods due to sensory sensitivities. Deviation from these preferences led to anxiety or physical reactions like gagging. Within the second focus group, these behaviors were more intense, such as depending on supplemental nutrition because of severe anxiety (P6). This underscores the different severity levels of selective eating behaviors and the necessity for customized interventions.
- **External Situations:** Maintaining consistent eating routines in external environments, such as schools or social gatherings, proved difficult. Parents reported that external caregivers often failed to recognize the child's specific needs, leading to setbacks. *"At home you have influence, but when your child goes to school, others have their own opinions, which can ruin everything."* – P4.

In addition to thematic insights, empathy maps (Figures 1) were created to illustrate the emotional and cognitive dimensions of both parents and children. These maps offered a structured visualization

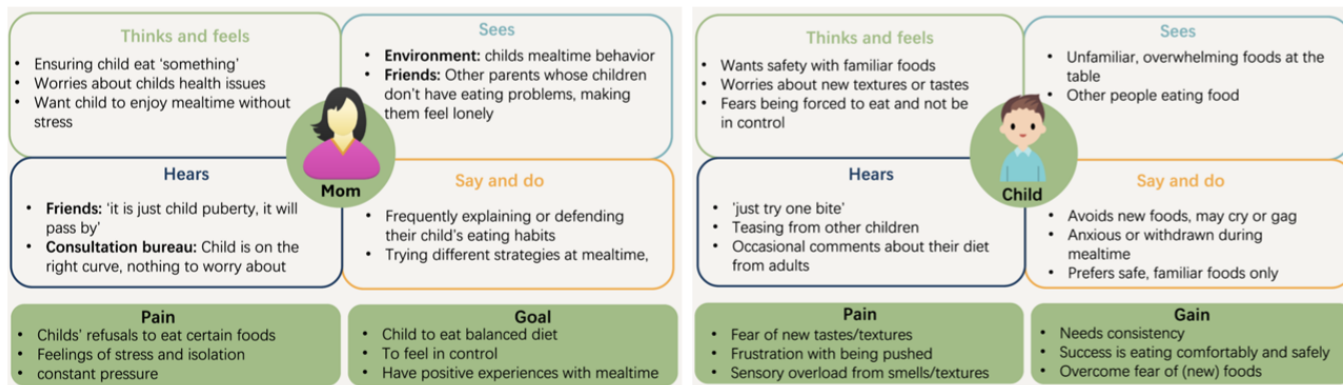


Figure 1: Empathy map of the parent (mothers) and the child respectively

of their experiences, detailing what they think, feel, see, hear, say, and do, along with their pains and goals. These maps informed the design by emphasizing the unique needs of both groups.

4.2 Effective and Ineffective Strategies

Through thematic analysis, we identified three core strategies that parents find effective in encouraging their children to try new foods. These strategies primarily revolve around maintaining consistency, fostering engagement, and reducing external pressure. At the same time, our analysis highlighted several ineffective strategies that often lead to heightened food aversion or increased mealtime stress. Understanding these pitfalls helps clarify what to avoid when designing interventions and support systems for parents.

Effective Strategies. One of the most successful approaches parents reported was consistency and structure. Gradual, step-by-step exposure to new foods, combined with predictable mealtime routines, played a crucial role in reducing children's anxiety and increasing their willingness to try unfamiliar foods. By maintaining a structured approach, parents found that children gradually became more comfortable with novel tastes and textures, making mealtimes less stressful and more predictable. Another key strategy was engagement and motivation, where parents emphasized the importance of creating a positive, low-pressure environment during meals. A calm atmosphere, coupled with encouragement in the form of praise or small rewards, helped children associate mealtimes with positive experiences. Parents noted that when children felt in control and received gentle reinforcement, they showed greater openness to trying new foods. Finally, reducing pressure emerged as an essential factor in supporting children's willingness to explore different foods. Parents found that setting clear boundaries while minimizing external pressures—such as rushing meals or receiving unsolicited advice from others—helped reduce resistance and anxiety. However, experiences varied among focus group participants, with some reporting that simply extending the duration of meals was ineffective in alleviating pressure. In such cases, prolonged exposure without additional supportive strategies led to increased stress rather than improved eating behaviors.

Ineffective Strategies. In contrast, parents identified several strategies that proved counterproductive. External pressure, such

as stressful meal environments, time constraints, or societal expectations, often undermined children's willingness to try new foods. One parent described how the short lunch breaks at school created significant mealtime stress, making it difficult for children to eat in a relaxed manner: "... it's really just at school, especially between noon, that's really drama. It is indeed way too short and way too much to do." – P2. Another ineffective approach involved inappropriate strategies such as punishments or abrupt, large changes in dietary expectations. Parents who attempted to enforce strict rules or use punitive measures, such as withholding food altogether, found that these tactics exacerbated food refusal rather than encouraging dietary expansion. One participant recounted a particularly severe experience: "One time I said 'then don't eat anything. After two weeks, he was dehydrated because he did not eat at all.'" – P3. Finally, sensory challenges were a recurring issue that contributed to food refusal. Overwhelming stimuli, such as noisy or brightly lit dining environments, created sensory overload, making it difficult for children to focus on eating. Parents observed that highly stimulating settings heightened aversion and resistance to new foods, reinforcing negative associations with mealtime.

By understanding both the effective and ineffective strategies employed by parents, it becomes clear that supportive, structured, and pressure-free approaches yield the best outcomes. These insights inform the design of interventions aimed at fostering positive eating behaviors in children while avoiding common pitfalls.

4.3 Exploring New Solutions

In the third round, parents participated in a dot-voting exercise to identify key features of a parent app and the game-based platform for children. Following the vote, further ideas were presented during an open discussion. Their recommendations comprised:

• Parent App Preferences:

- Tools for tracking sensory-related behaviors, food preferences, and progress milestones.
- Access to expert guidance and peer support for validation and shared experiences.
- Motivational feedback to reduce parental stress.
- Cost-effective solutions were a priority, and while parents remained cautiously optimistic about AI integration, they

highlighted the need for solutions that could adapt to their specific needs.

- **Children's Game Features:**

- Hybrid game structure: A mix of digital and physical elements, as some children might initially fear real food interactions.
- Rewards and collaboration: Rewards were expected to be effective in motivating children, and collaborative or competitive play was thought to add engagement. However, a parent noted that competitive play would only work if the child experienced success, as failure could lead to frustration or tantrums.
- Non-open-ended games: Provide structure and consistency that align with the needs of children with ASD.
- Aesthetic aspects: Parents also preferred energizing games with expressive colors to boost engagement, although this preference appeared to contradict the desire for neutral, calming elements during mealtimes. This indicates that sensory needs may change depending on activity (e.g., play versus eating).

- **Additional Ideas:**

- Structured and Progressive Challenges: Progressive challenges and adaptable difficulty levels were seen as essential to maintain engagement. As one parent noted, *"...if my child has been doing this game for some time, they might choose to take the actual physical food and work with that"* (P5).
- Engagement and Motivation: Positive reinforcement and personalization (e.g., tailoring rewards to the child's progress) were highlighted as crucial design elements.
- Sensory Comfort and Anxiety Reduction: Using interactive storytelling and visual cues to engage the child within the game was considered an effective strategy to reduce anxiety.

4.4 Design Brief

Following the dot-voting exercise, participants completed the second questionnaire on their expectations of a sensory game-based tool and its companion app. Thematic analysis identified four main themes that shape parental expectations: Parental Support and Insights, Engagement and Adaptability, Motivational and Emotional Support, and Compatibility with Existing Tools and Therapies. The highest-rated theme, Parental Support and Insights (3.92 ± 1.12) highlights the importance of structured guidance and features that assist parents in managing their child's sensory experiences. Engagement and Adaptability (3.46 ± 1.09) was also emphasized, with parents valuing interactive and customizable elements. However, aspects related to Motivational and Emotional Support (3.24 ± 1.27) and Seamless Integration (3.27 ± 1.28) with existing therapeutic tools emerged as areas for further development. To translate these insights into actionable design principles, a design brief was developed to guide the future implementation of the platform (Figure 2). The system consists of two core components: a companion app for parents and an interactive game for children, each addressing key challenges related to sensory sensitivities and picky eating behaviors. The parent app should provide structured guidance, progress

tracking, and personalized sensory adjustments. Motivational feedback should support parents under pressure, helping them navigate mealtime challenges. Real-time tracking should visualize a child's adaptation to sensory experiences using pictograms and checklists, while adaptive algorithms tailor recommendations based on behavior. A community support feature should connect parents with peer groups for shared strategies. The children's game should be a sensory-adaptive experience encouraging gradual food exposure through interactive storytelling. Children should be able to choose food-related adventures, reinforcing familiarity with textures and flavors. Multi-sensory cues (animations, sounds, haptic feedback) should gently introduce sensory experiences, while contextual food exposure links in-game events to real-life meals, prompting parental involvement. To support parent-child collaboration, the game should include guided interactions, such as describing food properties or engaging in co-play activities like virtual cooking. A feedback loop should let parents adjust difficulty levels, ensuring challenges remain appropriate. For sensory flexibility, children should be able to adjust textures, colors, and sounds in-game. Multiple interaction modes should offer touch-based exploration or sound cues, while gradual sensory exposure enables progression from familiar to novel foods at a comfortable pace. A progressive difficulty model should guide children from visual familiarity to virtual tasting and real-world interactions. Adaptive learning should track engagement, dynamically adjusting difficulty. Optional challenges should let children engage with food elements at their own pace. A reward system should reinforce positive eating behaviors. Children should earn tokens or points to unlock stickers, avatars, or special content. Personalized encouragement messages should celebrate progress (e.g., "You explored a new texture today—great job!"). Real-life rewards should also be integrated, encouraging parents to introduce small tangible incentives like a fun mealtime activity. By structuring sensory exposure in an adaptive, interactive, and parent-inclusive way, the proposed design aims to create an engaging and behaviorally effective intervention for both parents and children.

5 Discussion

This study contributes new insights into the development of sensory-based digital interventions for selective eating in children with ASD by identifying effective and ineffective parental strategies and translating them into digital design requirements. While previous research has explored selective eating in children with ASD, it has primarily focused on clinical interventions, behavioral therapy, or child-only digital tools. In contrast, this study highlights the role of parents as co-designers, demonstrating how digital interventions can support both children and families in managing eating challenges. Further, digital interventions tailored to family needs have been shown to improve adherence and long-term eating behavior outcomes in children with ASD [26].

Current clinician-led approaches, such as ABA, often treat parents as facilitators rather than adaptive decision-makers in real-world contexts. However, this study reveals that parents rely on flexible, trial-and-error strategies tailored to their child's sensory needs, making rigid intervention models less effective. The findings emphasize the necessity of adaptable digital tools that reflect real-life

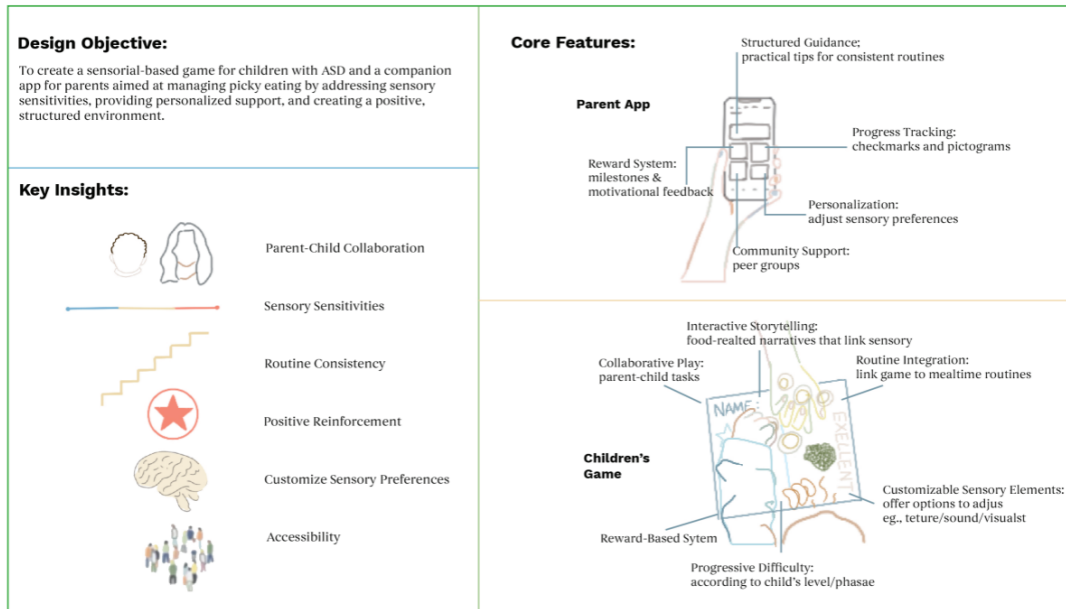


Figure 2: Design brief

problem-solving approaches rather than enforcing strict behavioral reinforcement. Unlike existing ASD-related digital tools that primarily focus on tracking behaviors and reinforcement mechanisms, this approach prioritizes sensory-driven, hybrid digital-physical experiences that offer personalized, parent-guided engagement.

Many existing digital interventions, such as Tiny Tastes and Food Dudes, incorporate reward-based gamification, which has been shown to encourage food acceptance in children. However, these approaches often fail to account for sensory sensitivities and behavioral rigidity common in ASD. While reinforcement-based strategies can be effective in increasing dietary variety when properly individualized [18], they often fail to account for sensory sensitivities and behavioral rigidity common in ASD. Our findings highlight the importance of gradual sensory exposure and adaptive engagement mechanisms, ensuring a more inclusive experience. Rather than relying solely on extrinsic motivation through tokens and rewards, this study emphasizes intrinsic motivation through sensory-based engagement, including interactive storytelling and structured but adaptable food exposure challenges. The use of non-competitive mechanics reduces anxiety while maintaining engagement, ensuring children are not pressured into progression. Additionally, hybrid digital-physical interactions create a bridge between virtual play and real-world food experiences. Research supports this transition model, as multi-sensory gradual exposure has been shown to improve food acceptance in children with ASD [6]. Unlike existing digital tools that rely on immediate rewards, this approach aligns with best practices in sensory-based feeding therapy, allowing children to engage at their own pace. Professional feeding therapies often have long waiting periods of up to 12 months in the Netherlands and are costly, leaving parents to manage feeding challenges independently. By integrating evidence-based techniques such as food chaining, structured exposure, and real-time monitoring, this

intervention provides a cost-effective, home-based alternative to traditional therapy.

Despite its contributions, this study has several limitations. The small sample size of six participants, all mothers, restricts the generalizability of the findings. Additionally, conducting the focus group online and dividing it into two smaller groups may have limited the depth of interaction and discussion. Furthermore, the study did not address the needs of children with diverse sensory profiles, highlighting the need for future research with more inclusive samples. By addressing these areas, future research can ensure that the proposed design is scientifically robust and practically effective, ultimately supporting children with ASD and their families in a meaningful and sustainable way.

6 Conclusion

This study highlights the potential of a game-based intervention as a viable and accessible support solution for selective eating in children with ASD and their parents. Unlike prior work that focuses on child-centered interventions, this research highlights the essential role of parents as co-designers and decision-makers. The findings contribute empirical insights into the effective and ineffective strategies to manage children with selective eating problem. By incorporating gradual sensory exposure, positive reinforcement, and supporting parents through educational resources and community features, the proposed design has the potential to create meaningful and positive changes for both children and their families. This design brief is grounded in the ABA and OT frameworks, which provide evidence-based strategies to address selective eating behaviors. Moreover, it aligns with the theoretical models while adapting them for digital delivery. This approach not only bridges research and practice but also addresses the practical realities faced by families, offering a flexible and accessible intervention.

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A Questionnaires

A.1 Demographic and Use of technology

Questions About Yourself

- What is your age group?
 - (1) 18-24
 - (2) 25-34
 - (3) 35-44
 - (4) 45-54
 - (5) 55-64
 - (6) 65 years or older
- Gender:
 - (1) Male
 - (2) Female
 - (3) Non-binary/Other
 - (4) Prefer not to say
- Number of children:
 - (1) 1
 - (2) 2
 - (3) 3
 - (4) 4 or more
- Educational level:
 - (1) Less than high school
 - (2) High school diploma or equivalent (e.g., secondary vocational education, senior general secondary education, pre-university education)
 - (3) Some higher education
 - (4) Associate degree or vocational diploma
 - (5) Bachelor's degree (applied sciences or university)
 - (6) Master's degree or professional qualification (e.g., university master's, applied sciences master's, or another professional diploma)
- Employment status:
 - (1) Full-time employed
 - (2) Part-time employed
 - (3) Self-employed
 - (4) Unemployed
 - (5) Homemaker
 - (6) Retired
 - (7) Student
 - (8) Other (please specify)

Questions About Your Child

- What is your child's age?
- What is your child's gender?
 - (1) Male
 - (2) Female
 - (3) Non-binary/Other
 - (4) Prefer not to say
- Autism Spectrum Disorder (ASD) Diagnosis Information:
 - (1) Before the age of 2 years
 - (2) Between the ages of 2-4 years
 - (3) Between the ages of 5-8 years
- Your child's eating behavior:
 - (1) Very picky (limited to a few foods)
 - (2) Moderately picky
 - (3) Occasionally picky
 - (4) Not picky
- Use of technology:

- (1) Do you currently use apps, websites, or digital tools to support your child's healthcare or sensory needs?
 - (a) Yes (please specify)
 - (b) No
- (2) How often do you use digital tools (apps, websites) to support your child's needs?
 - (a) Daily
 - (b) Weekly
 - (c) Monthly
 - (d) Rarely
 - (e) Never
- (3) How comfortable are you using digital tools (apps, websites) for your child's care?
 - (a) Very comfortable
 - (b) Comfortable
 - (c) Neutral
 - (d) Uncomfortable
 - (e) Very uncomfortable

A.2 Expectation Scale for a Sensory Game-Based Tool and Companion App

Please indicate the extent to which you agree or disagree with the following statements regarding a potential sensory play-based tool for children with ASD and the accompanying app for parents, designed to assist with eating behavior and sensory needs.

Use the following scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree

- (1) I believe the sensory play-based tool will be able to understand my child's sensory preferences.
- (2) I believe the sensory play-based tool will be more than just a simple game.
- (3) I believe the sensory play-based tool will anticipate my child's responses to certain sensory stimuli.
- (4) I believe my child will engage with the sensory play-based tool in a meaningful way.
- (5) I believe the sensory play-based tool will be comparable to other advanced games my child has previously used.
- (6) I believe the sensory play-based tool will understand my child's emotional responses during gameplay.
- (7) I believe the sensory play-based tool will recognize when my child is engaged or losing interest.
- (8) I believe the sensory play-based tool will provide emotional support to my child during meals or sensory activities.
- (9) I believe the sensory play-based tool will help calm my child when they feel anxious about eating or sensory experiences.
- (10) I believe the accompanying parent app will enable me to monitor and track my child's progress over time.
- (11) I believe the accompanying parent app will provide personalized recommendations based on my child's gameplay and eating habits.
- (12) I believe the accompanying parent app will suggest activities or exercises to enhance my child's sensory experiences related to food.
- (13) I believe the sensory play-based tool will make mealtimes more enjoyable and engaging for my child.
- (14) I believe the sensory play-based tool will help my child feel more comfortable trying new foods.
- (15) I believe the accompanying parent app will provide feedback on my child's sensory preferences and eating behavior.
- (16) I believe the sensory play-based tool and the accompanying app will work effectively alongside other therapies my child receives.
- (17) I believe the accompanying parent app will help me better understand my child's sensory needs related to food.
- (18) I believe the sensory play-based tool and the accompanying app will adapt to my child's changing needs over time.
- (19) I believe the sensory play-based tool will be fun and engaging for my child to use.
- (20) I believe the sensory play-based tool and the accompanying app will become an integral part of our daily routine.

B Dot-voting Schema

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Needs and Support for Parents Guidance					
WHY					
Practical information	Medical guidance	Psychological counseling	tips & tricks		
BY WHOM					
Medical specialist	AI Agent	Other parents	Family & friends	No one	
HOW					
Forum	Chat	Call	Video calling	In-person	Other
WHEN					
Before intervention	During intervention	After intervention	In free time		
INPUT					
Tracking eating preferences	Tracking sensory-related behavior	Tracking Milestone	Setting goals		
OUTPUT					
Information	Understanding the cause of behavior	Motivational feedback	Progress report	Celebrating achievements	Suggestions for positive reinforcement
INVOLVEMENT IN INTERVENTION					
child does it alone, without parent participation		with a little support from the parent		Parent participation (playing together)	
accessible, affordable product			High quality, more expensive product		

Figure 3: Dot-voting schema - Design aspects for parental support

Child's Game			
Sensory Aspects Included in the Game Intervention			
visual (e.g., seeing food in different colors)	auditory (e.g., hearing from the sounds of food while eating/playing with it)	tactile (e.g., touching or playing with food)	olfactory (e.g., tasting food without smelling it)
olfactory (e.g., taste food without smelling it)	gustative (e.g., the taste of food adaptation/combining)	vestibular (e.g., the sound food that comes into contact with cutlery)	proprioception (e.g., crushing food)
ENVIRONMENT			
permanent place		removable	
GAME STRUCTURE			
physical	digital	hybrid	
Individual play	cooperate	competition	
reward		punish	
PREFERRED IN DESIGN			
Neutral colors		Expressive colours	
energetic		calming	
open-ended play form		closed play	

Figure 4: Dot-voting schema - Design aspect for child